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Branch: BE-CSE-AIML

Semester: 5

Subject: ADBMS

Experiment-06

Aim:

(A) Problem Statement

A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges.

Generate a daily sales comparison report showing the difference between the number of apples sold and the number of oranges sold on the same day. The difference should be calculated as:

$\text{Difference} = \text{Apples Sold} - \text{Oranges Sold}$

Display the sale date and the calculated difference. The result should be sorted by sale date in ascending order.

(B) Problem Statement

A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where:

- id represents the current node.
- p_id represents the parent node.

Classify every node into one of the following categories:

- Root: A node that does not have any parent.
- Leaf: A node that has a parent but does not have any children.
- Inner: A node that has both a parent and at least one child.

Display the node ID along with its category. Return the result sorted by node ID.

Tools / Platform Used:

- Windows
- PostgreSQL
- PgAdmin

Code and Output:

A.

```
1  CREATE TABLE Sales
2  (
3      sale_date DATE,
4      fruit VARCHAR(20),
5      sold_num INT,
6
7      PRIMARY KEY (sale_date, fruit)
8  );
9
10 INSERT INTO Sales
11 VALUES
12 ('2020-05-01','apples',10),
13 ('2020-05-01','oranges',8),
14
15 ('2020-05-02','apples',15),
16 ('2020-05-02','oranges',15),
17
18 ('2020-05-03','apples',20),
19 ('2020-05-03','oranges',0),
20
21 ('2020-05-04','apples',15),
22 ('2020-05-04','oranges',16);
23
```

```
24 SELECT sale_date,sum(
25     case
26         when fruit= 'apples' then sold_num
27         else -sold_num
28     end
29 )
30 from Sales
31 group by sale_date
32 order by sale_date;
```

Data Output Messages Notifications

	sale_date date	sum bigint
1	2020-05-01	2
2	2020-05-02	0
3	2020-05-03	20
4	2020-05-04	-1

B.

```
1  CREATE TABLE Tree (  
2      id INT,  
3      p_id INT  
4  );  
5  
6  INSERT INTO Tree (id, p_id) VALUES  
7  (1, NULL),  
8  (2, 1),  
9  (3, 1),  
10 (4, 2),  
11 (5, 2);  
12  
13 SELECT  
14     id,  
15     CASE  
16         WHEN p_id IS NULL THEN 'Root'  
17         WHEN id IN (SELECT p_id FROM Tree) THEN 'Inner'  
18         ELSE 'Leaf'  
19     END AS Type  
20 FROM Tree  
21 ORDER BY id;
```

Data Output Messages Notifications



	id integer	type text
1	1	Root
2	2	Inner
3	3	Leaf
4	4	Leaf
5	5	Leaf